3.9 THE CONTROLLED HALF-WAVE RECTIFIER

The half-wave rectifiers analyzed previously in this chapter are classified as uncontrolled rectifiers. Once the source and load parameters are established, the dc level of the output and the power transferred to the load are fixed quantities.

A way to control the output of a half-wave rectifier is to use an SCR[1] instead of a diode. Figure 3-13*a* shows a basic controlled half-wave rectifier with a resistive load. Two conditions must be met before the SCR can conduct:

1. The SCR must be forward-biased ($v_{SCR} > 0$).
2. A current must be applied to the gate of the SCR.

Unlike the diode, the SCR will not begin to conduct as soon as the source becomes positive. Conduction is delayed until a gate current is applied, which is the basis for using the SCR as a means of control. Once the SCR is conducting, the gate current can be removed and the SCR remains on until the current goes to zero.

Resistive Load

Figure 3-13*b* shows the voltage waveforms for a controlled half-wave rectifier with a resistive load. A gate signal is applied to the SCR at $\omega t = \alpha$, where α is the delay angle. The average (dc) voltage across the load resistor in Fig. 3-13*a* is

$$\boxed{V_o = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{2\pi}(1 + \cos\alpha)} \tag{3-52}$$

The power absorbed by the resistor is V_{rms}^2/R, where the rms voltage across the resistor is computed from

$$\begin{aligned} V_{rms} &= \sqrt{\frac{1}{2\pi}\int_{0}^{2\pi} v_o^2(\omega t)\, d(\omega t)} \\ &= \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi} [V_m \sin(\omega t)]^2\, d(\omega t)} \\ &= \frac{V_m}{2}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \end{aligned} \tag{3-53}$$

[1] Switching with other controlled turn-on devices such as transistors or IGBTs can be used to control the output of a converter.

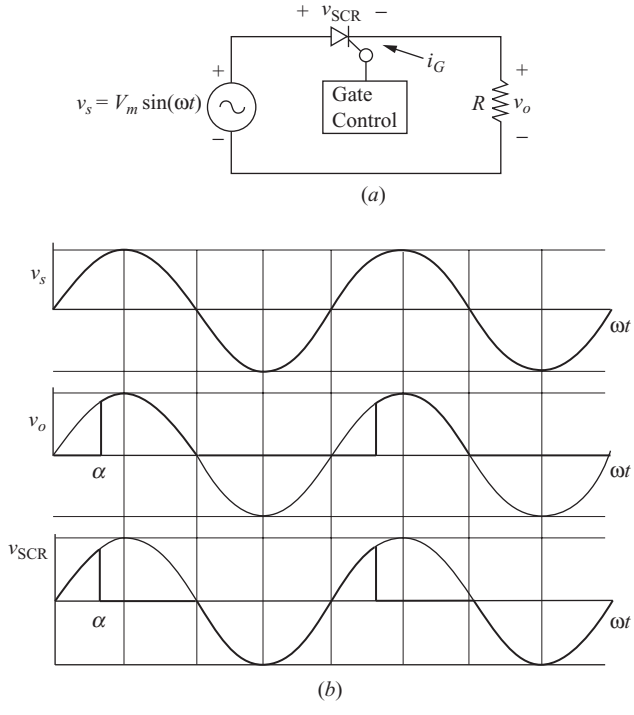


Figure 3-13 (*a*) A basic controlled rectifier; (*b*) Voltage waveforms.

EXAMPLE 3-10

Controlled Half-Wave Rectifier with Resistive Load

Design a circuit to produce an average voltage of 40 V across a 100-Ω load resistor from a 120-V rms 60-Hz ac source. Determine the power absorbed by the resistance and the power factor.

■ Solution

If an uncontrolled half-wave rectifier is used, the average voltage will be $V_m/\pi = 120\sqrt{2}/\pi = 54$ V. Some means of reducing the average resistor voltage to the design specification of 40 V must be found. A series resistance or inductance could be added to an uncontrolled rectifier, or a controlled rectifier could be used. The controlled rectifier of Fig. 3-13*a* has the advantage of not altering the load or introducing losses, so it is selected for this application.

Equation (3-52) is rearranged to determine the required delay angle:

$$\alpha = \cos^{-1}\left[V_o\left(\frac{2\pi}{V_m}\right) - 1\right]$$

$$= \cos^{-1}\left\{40\left[\frac{2\pi}{\sqrt{2}(120)}\right] - 1\right\} = 61.2° = 1.07 \text{ rad}$$

Equation (3-53) gives

$$V_{\text{rms}} = \frac{\sqrt{2}(120)}{2}\sqrt{1 - \frac{1.07}{\pi} + \frac{\sin[2(1.07)]}{2\pi}} = 75.6 \text{ V}$$

Load power is

$$P_R = \frac{V_{\text{rms}}^2}{R} = \frac{(75.6)^2}{100} = 57.1 \text{ W}$$

The power factor of the circuit is

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{S,\text{rms}} I_{\text{rms}}} = \frac{57.1}{(120)(75.6/100)} = 0.63$$

3.3 RESISTIVE-INDUCTIVE LOAD

Industrial loads typically contain inductance as well as resistance. As the source voltage goes through zero, becoming positive in the circuit of Fig. 3-2*a*, the diode becomes forward-biased. The Kirchhoff voltage law equation that describes the current in the circuit for the forward-biased ideal diode is

$$V_m \sin(\omega t) = Ri(t) + L\frac{di(t)}{dt} \tag{3-4}$$

The solution can be obtained by expressing the current as the sum of the forced response and the natural response:

$$i(t) = i_f(t) + i_n(t) \tag{3-5}$$

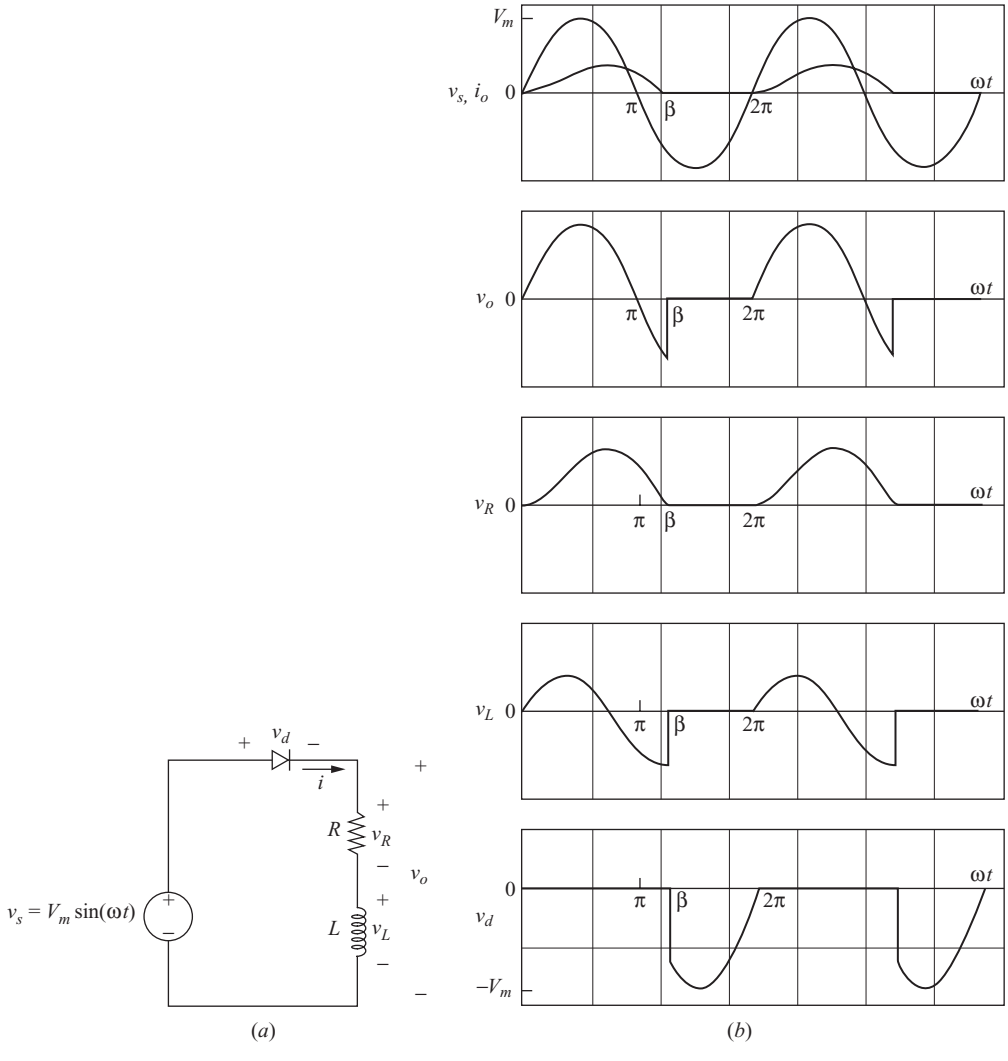


Figure 3-2 (*a*) Half-wave rectifier with an *RL* load; (*b*) Waveforms.

The forced response for this circuit is the current that exists after the natural response has decayed to zero. In this case, the forced response is the steady-state sinusoidal current that would exist in the circuit if the diode were not present. This steady-state current can be found from phasor analysis, resulting in

$$i_f(t) = \frac{V_m}{Z} \sin(\omega t - \theta) \tag{3-6}$$

where $Z = \sqrt{R^2 + (\omega L)^2}$ and $\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$

The natural response is the transient that occurs when the load is energized. It is the solution to the homogeneous differential equation for the circuit without the source or diode:

$$R\,i(t) + L\,\frac{di(t)}{dt} = 0 \tag{3-7}$$

For this first-order circuit, the natural response has the form

$$i_n(t) = Ae^{-t/\tau} \tag{3-8}$$

where τ is the time constant L/R and A is a constant that is determined from the initial condition. Adding the forced and natural responses gets the complete solution.

$$i(t) = i_f(t) + i_n(t) = \frac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-t/\tau} \tag{3-9}$$

The constant A is evaluated by using the initial condition for current. The initial condition of current in the inductor is zero because it was zero before the diode started conducting and it cannot change instantaneously.

Using the initial condition and Eq. (3-9) to evaluate A yields

$$\begin{aligned} i(0) &= \frac{V_m}{Z}\sin(0 - \theta) + Ae^0 = 0 \\ A &= -\frac{V_m}{Z}\sin(-\theta) = \frac{V_m}{Z}\sin\theta \end{aligned} \tag{3-10}$$

Substituting for A in Eq. (3-9) gives

$$\begin{aligned} i(t) &= \frac{V_m}{Z}\sin(\omega t - \theta) + \frac{V_m}{Z}\sin(\theta)e^{-t/\tau} \\ &= \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-t/\tau}\right] \end{aligned} \tag{3-11}$$

It is often convenient to write the function in terms of the angle ωt rather than time. This merely requires ωt to be the variable instead of t. To write the above equation in terms of angle, t in the exponential must be written as ωt, which requires τ to be multiplied by ω also. The result is

$$i(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\omega t/\omega\tau}\right] \tag{3-12}$$

A typical graph of circuit current is shown in Fig. 3-2*b*. Equation (3-12) is valid for positive currents only because of the diode in the circuit, so current is zero when the function in Eq. (3-12) is negative. When the source voltage again becomes positive, the diode turns on, and the positive part of the waveform in Fig. 3-2*b* is repeated. This occurs at every positive half-cycle of the source. The voltage waveforms for each element are shown in Fig. 3-2*b*.

Note that the diode remains forward-biased longer than π rad and that the source is negative for the last part of the conduction interval. This may seem

unusual, but an examination of the voltages reveals that Kirchhoff's voltage law is satisfied and there is no contradiction. Also note that the inductor voltage is negative when the current is decreasing ($v_L = L\,di/dt$).

The point when the current reaches zero in Eq. (3-12) occurs when the diode turns off. The first positive value of ωt in Eq. (3-12) that results in zero current is called the extinction angle β.

Substituting $\omega t = \beta$ in Eq. (3-12), the equation that must be solved is

$$i(\beta) = \frac{V_m}{Z}\left[\sin(\beta - \theta) + \sin(\theta)e^{-\beta/\omega\tau}\right] = 0 \tag{3-13}$$

which reduces to

$$\boxed{\sin(\beta - \theta) + \sin(\theta)e^{-\beta/\omega\tau} = 0} \tag{3-14}$$

There is no closed-form solution for β, and some numerical method is required. To summarize, the current in the half-wave rectifier circuit with *RL* load (Fig. 3-2) is expressed as

$$i(\omega t) = \begin{cases} \dfrac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\omega t/\omega\tau}\right] & \text{for } 0 \le \omega t \le \beta \\ 0 & \text{for } \beta \le \omega t \le 2\pi \end{cases} \tag{3-15}$$

where $Z = \sqrt{R^2 + (\omega L)^2}$ $\theta = \tan^{-1}\left(\dfrac{\omega L}{R}\right)$ and $\tau = \dfrac{L}{R}$

The average power absorbed by the load is $I_{\text{rms}}^2 R$, since the average power absorbed by the inductor is zero. The rms value of the current is determined from the current function of Eq. (3-15).

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i^2(\omega t)\,d(\omega t)} = \sqrt{\frac{1}{2\pi}\int_0^{\beta} i^2(\omega t)\,d(\omega t)} \tag{3-16}$$

Average current is

$$I_o = \frac{1}{2\pi}\int_0^{\beta} i(\omega t)\,d(\omega t) \tag{3-17}$$

EXAMPLE 3-2

Half-Wave Rectifier with *RL* Load

For the half-wave rectifier of Fig. 3-2*a*, $R = 100\ \Omega$, $L = 0.1$ H, $\omega = 377$ rad/s, and $V_m = 100$ V. Determine (*a*) an expression for the current in this circuit, (*b*) the average current, (*c*) the rms current, (*d*) the power absorbed by the *RL* load, and (*e*) the power factor.

■ **Solution**

For the parameters given,

$$Z = [R^2 + (\omega L)^2]^{0.5} = 106.9\,\Omega$$
$$\theta = \tan^{-1}(\omega L/R) = 20.7^\circ = 0.361 \text{ rad}$$
$$\omega t = \omega L/R = 0.377 \text{ rad}$$

(*a*) Equation (3-15) for current becomes

$$i(\omega t) = 0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377} \quad \text{A} \quad \text{for } 0 \le \omega t \le \beta$$

Beta is found from Eq. (3-14).

$$\sin(\beta - 0.361) + \sin(0.361)e^{-\beta/0.377} = 0$$

Using a numerical root-finding program, β is found to be 3.50 rad, or 201°

(*b*) Average current is determined from Eq. (3-17).

$$I_o = \frac{1}{2\pi}\int_0^{3.50}\left[0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377}\right]d(\omega t) = 0.308 \text{ A}$$

(A numerical integration program is recommended.)

(*c*) The rms current is found from Eq. (3-16) to be

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{3.50}\left[0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377}\right]^2 d(\omega t)} = 0.474 \text{ A}$$

(*d*) The power absorbed by the resistor is

$$P = I_{\text{rms}}^2 R = (0.474)^2(100) = 22.4 \text{ W}$$

The average power absorbed by the inductor is zero. Also P can be computed from the definition of average power:

$$P = \frac{1}{2\pi}\int_0^{2\pi} p(\omega t)\,d(\omega t) = \frac{1}{2\pi}\int_0^{2\pi} v(\omega t)i(\omega t)\,d(\omega t)$$
$$= \frac{1}{2\pi}\int_0^{3.50}[100\sin(\omega t)]\left[0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377}\right]d(\omega t)$$
$$= 22.4 \text{ W}$$

(*e*) The power factor is computed from the definition pf = P/S, and P is power supplied by the source, which must be the same as that absorbed by the load.

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{\text{rms}}} = \frac{22.4}{(100/\sqrt{2})0.474} = 0.67$$

Note that the power factor is *not* cos θ.

3.7 THE FREEWHEELING DIODE

Creating a DC Current

A freewheeling diode, D_2 in Fig. 3-7*a*, can be connected across an *RL* load as shown. The behavior of this circuit is somewhat different from that of the half-wave rectifier of Fig. 3-2. The key to the analysis of this circuit is to determine when each diode conducts. First, it is observed that both diodes cannot be forward-biased at the same time. Kirchhoff's voltage law around the path containing the source and the two diodes shows that one diode must be reverse-biased. Diode D_1 will be on when the source is positive, and diode D_2 will be on when the source is negative.

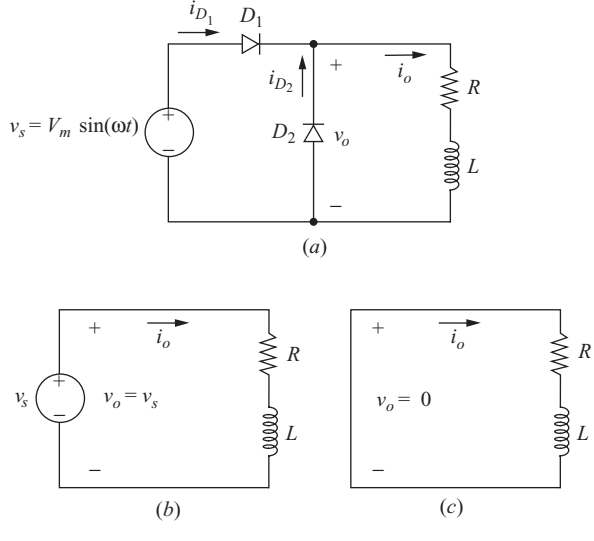


Figure 3-7 (*a*) Half-wave rectifier with freewheeling diode; (*b*) Equivalent circuit for $v_s > 0$; (*c*) Equivalent circuit for $v_s < 0$.

For a positive source voltage,

- D_1 is on.
- D_2 is off.
- The equivalent circuit is the same as that of Fig. 3-2, shown again in Fig. 3-7*b*.
- The voltage across the *RL* load is the same as the source.

For a negative source voltage,

- D_1 is off.
- D_2 is on.
- The equivalent circuit is the same at that of Fig. 3-7*c*.
- The voltage across the *RL* load is zero.

Since the voltage across the *RL* load is the same as the source voltage when the source is positive and is zero when the source is negative, the load voltage is a half-wave rectified sine wave.

When the circuit is first energized, the load current is zero and cannot change instantaneously. The current reaches periodic steady state after a few periods (depending on the L/R time constant), which means that the current at the end of a period is the same as the current at the beginning of the period, as shown in Fig. 3-8. The steady-state current is usually of greater interest than the transient that occurs when the circuit is first energized. Steady-state load, source, and diode currents are shown in Fig. 3-9.

The Fourier series for the half-wave rectified sine wave for the voltage across the load is

$$v(t) = \frac{V_m}{\pi} + \frac{V_m}{2}\sin(\omega_0 t) - \sum_{n=2,4,6\ldots}^{\infty} \frac{2V_m}{(n^2-1)\pi}\cos(n\omega_0 t) \qquad (3\text{-}34)$$

The current in the load can be expressed as a Fourier series by using superposition, taking each frequency separately. The Fourier series method is illustrated in Example 3-7.

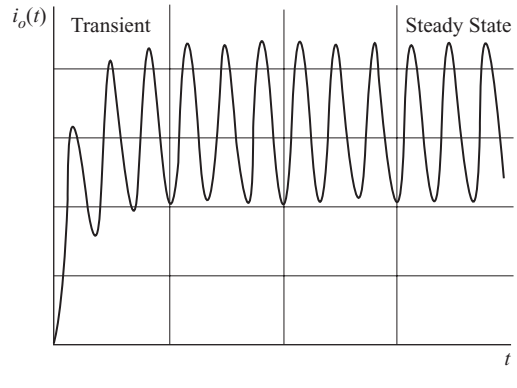


Figure 3-8 Load current reaching steady state after the circuit is energized.

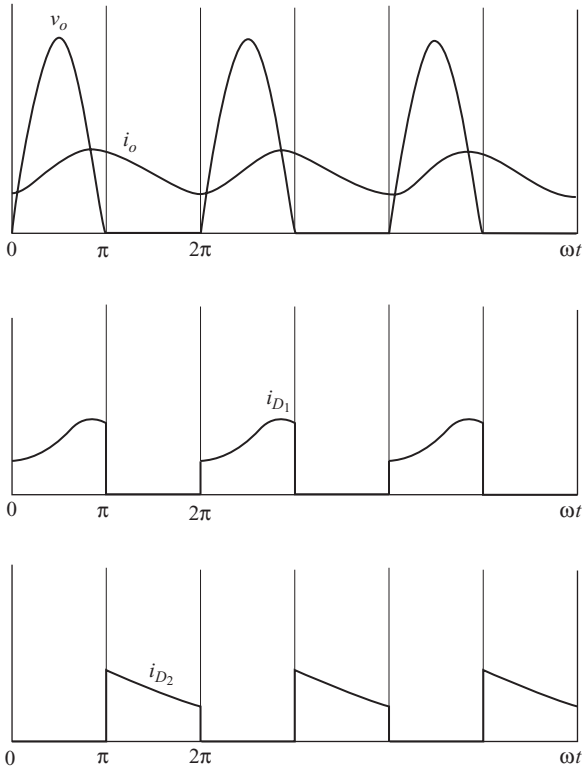


Figure 3-9 Steady-state load voltage and current waveforms with freewheeling diode.

EXAMPLE 3-7

Half-Wave Rectifier with Freewheeling Diode

Determine the average load voltage and current, and determine the power absorbed by the resistor in the circuit of Fig. 3-7*a*, where $R = 2\ \Omega$ and $L = 25$ mH, V_m is 100 V, and the frequency is 60 Hz.

■ Solution

The Fourier series for this half-wave rectified voltage that appears across the load is obtained from Eq. (3-34). The average load voltage is the dc term in the Fourier series:

$$V_o = \frac{V_m}{\pi} = \frac{100}{\pi} = 31.8 \text{ V}$$

Average load current is

$$I_o = \frac{V_o}{R} = \frac{31.8}{2} = 15.9 \text{ A}$$

Load power can be determined from $I_{\text{rms}}^2 R$, and rms current is determined from the Fourier components of current. The amplitudes of the ac current components are determined from phasor analysis:

$$I_n = \frac{V_n}{Z_n}$$

where $$Z_n = |R + jn\omega_0 L| = |2 + jn377(0.025)|$$

The ac voltage amplitudes are determined from Eq. (3-34), resulting in

$$V_1 = \frac{V_m}{2} = \frac{100}{2} = 50 \text{ V}$$

$$V_2 = \frac{2V_m}{(2^2 - 1)\pi} = 21.2 \text{ V}$$

$$V_4 = \frac{2V_m}{(4^2 - 1)\pi} = 4.24 \text{ V}$$

$$V_6 = \frac{2V_m}{(6^2 - 1)\pi} = 1.82 \text{ V}$$

The resulting Fourier terms are as follows:

n	V_n (V)	Z_n (Ω)	I_n (A)
0	31.8	2.00	15.9
1	50.0	9.63	5.19
2	21.2	18.96	1.12
4	4.24	37.75	0.11
6	1.82	56.58	0.03

The rms current is obtained using Eq. (2-64).

$$I_{\text{rms}} = \sqrt{\sum_{k=0}^{\infty} I_{k,\text{rms}}} \approx \sqrt{15.9^2 + \left(\frac{5.19}{\sqrt{2}}\right)^2 + \left(\frac{1.12}{\sqrt{2}}\right)^2 + \left(\frac{0.11}{\sqrt{2}}\right)^2} = 16.34 \text{ A}$$

Notice that the contribution to rms current from the harmonics decreases as n increases, and higher-order terms are not significant. Power in the resistor is $I_{\text{rms}}^2 R = (16.34)^2 2 = 534$ W.

EXAMPLE 3-8

Half-Wave Rectifier with Freewheeling Diode: $L/R \rightarrow \infty$

For the half-wave rectifier with a freewheeling diode and *RL* load as shown in Fig. 3-7*a*, the source is 240 V rms at 60 Hz and $R = 8\ \Omega$. (*a*) Assume *L* is infinitely large. Determine the power absorbed by the load and the power factor as seen by the source. Sketch v_o, i_{D_1}, and i_{D_2}. (*b*) Determine the average current in each diode. (*c*) For a finite inductance, determine *L* such that the peak-to-peak current is no more than 10 percent of the average current.

■ Solution

(*a*) The voltage across the *RL* load is a half-wave rectified sine wave, which has an average value of V_m/π. The load current is

$$i(\omega t) = I_o = \frac{V_o}{R} = \frac{V_m/\pi}{R} = \frac{\left(240\sqrt{2}\right)/\pi}{8} = 13.5\ \text{A} \approx I_{\text{rms}}$$

Power in the resistor is

$$P = (I_{\text{rms}})^2 R = (13.5)^2 8 = 1459\ \text{W}$$

Source rms current is computed from

$$I_{s,\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} (13.5)^2\, d(\omega t)} = 9.55\ \text{A}$$

The power factor is

$$\text{pf} = \frac{P}{V_{s,\text{rms}} I_{s,\text{rms}}} = \frac{1459}{(240)(9.55)} = 0.637$$

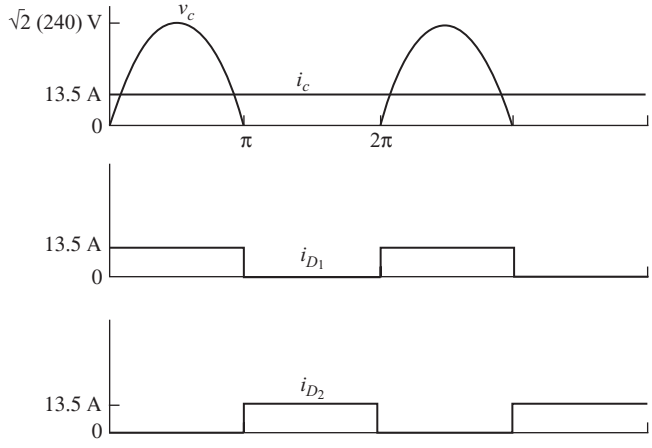


Figure 3-10 Waveforms for the half-wave rectifier with freewheeling diode of Example 3-8 with $L/R \rightarrow \infty$.

Voltage and current waveforms are shown in Fig. 3-10.

(*b*) Each diode conducts for one-half of the time. Average current for each diode is $I_o/2 = 13.5/2 = 6.75$ A.

(*c*) The value of inductance required to limit the variation in load current to 10 percent can be approximated from the fundamental frequency of the Fourier series. The voltage input to the load for $n = 1$ in Eq. (3-34) has amplitude $V_m/2 = \sqrt{2}(240)/2 = 170$ V the peak-to-peak current must be limited to

$$\Delta I_o = (0.10)(I_o) = (0.10)(13.5) = 1.35 \text{ A}$$

which corresponds to an amplitude of $1.35/2 = 0.675$ A. The load impedance at the fundamental frequency must then be

$$Z_1 = \frac{V_1}{I_1} = \frac{170}{0.675} = 251\ \Omega$$

The load impedance is

$$Z_1 = 251 = |R + j\omega L| = |8 + j377L|$$

Since the 8-Ω resistance is negligible compared to the total impedance, the inductance can be approximated as

$$L \approx \frac{Z_1}{\omega} = \frac{251}{377} = 0.67 \text{ H}$$

The inductance will have to be slightly larger than 0.67 H because Fourier terms higher than $n = 1$ were neglected in this estimate.

3.5 *RL*-SOURCE LOAD

Supplying Power to a DC Source from an AC Source

Another variation of the half-wave rectifier is shown in Fig. 3-5*a*. The load consists of a resistance, an inductance, and a dc voltage. Starting the analysis at $\omega t = 0$ and assuming the initial current is zero, recognize that the diode will remain off as long as the voltage of the ac source is less than the dc voltage. Letting α be the value of ωt that causes the source voltage to be equal to V_{dc},

$$V_m \sin \alpha = V_{dc}$$

or

$$\boxed{\alpha = \sin^{-1}\left(\frac{V_{dc}}{V_m}\right)} \qquad (3\text{-}18)$$

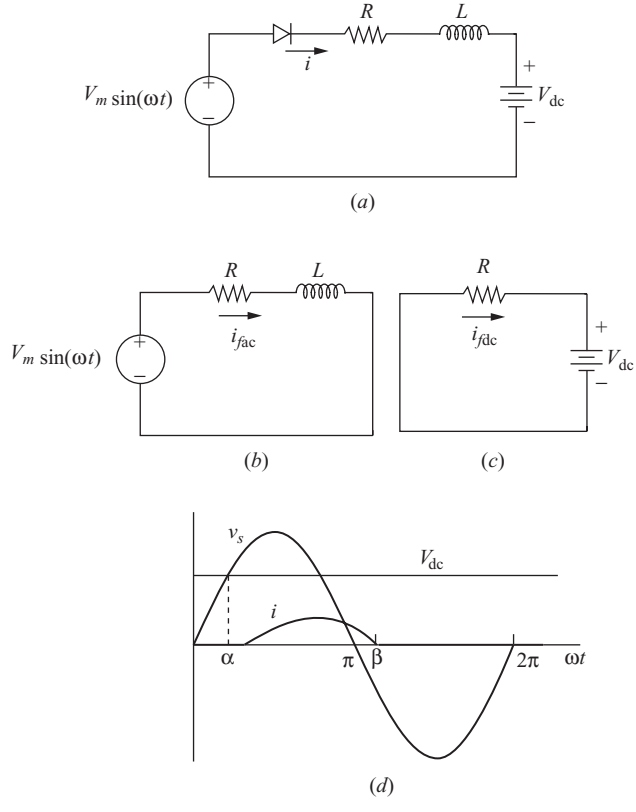


Figure 3-5 (*a*) Half-wave rectifier with *RL* source load; (*b*) Circuit for forced responce from ac source; (*c*) Circuit for forced responce from dc source; (*d*) Waveforms.

The diode starts to conduct at $\omega t = \alpha$. With the diode conducting, Kirchhoff's voltage law for the circuit yields the equation

$$V_m \sin(\omega t) = Ri(t) + L\frac{di(t)}{dt} + V_{dc} \tag{3-19}$$

Total current is determined by summing the forced and natural responses:

$$i(t) = i_f(t) + i_n(t)$$

The current $i_f(t)$ is determined using superposition for the two sources. The forced response from the ac source (Fig. 3-5*b*) is $(V_m/Z)\sin(\omega t - \theta)$. The forced response due to the dc source (Fig. 3-5*c*) is $-V_{dc}/R$. The entire forced response is

$$i_f(t) = \frac{V_m}{Z}\sin(\omega t - \theta) - \frac{V_{dc}}{R} \tag{3-20}$$

The natural response is

$$i_n(t) = Ae^{-t/\tau} \tag{3-21}$$

Adding the forced and natural responses gives the complete response.

$$i(\omega t) = \begin{cases} \dfrac{V_m}{Z}\sin(\omega t - \theta) - \dfrac{V_{dc}}{R} + Ae^{-\omega t/\omega\tau} & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases} \tag{3-22}$$

The extinction angle β is defined as the angle at which the current reaches zero, as was done earlier in Eq. (3-15). Using the initial condition of $i(\alpha) = 0$ and solving for A,

$$A = \left[-\frac{V_m}{Z}\sin(\alpha - \beta) + \frac{V_{\text{dc}}}{R}\right]e^{\alpha/\omega\tau} \tag{3-23}$$

Figure 3-5*d* shows voltage and current waveforms for a half-wave rectifier with *RL*-source load.

The average power absorbed by the resistor is $I_{\text{rms}}^2 R$, where

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\beta} i^2(\omega t)\, d(\omega t)} \tag{3-24}$$

The average power absorbed by the dc source is

$$P_{\text{dc}} = I_o V_{\text{dc}} \tag{3-25}$$

where I_o is the average current, that is,

$$I_o = \frac{1}{2\pi}\int_{\alpha}^{\beta} i(\omega t)\, d(\omega t) \tag{3-26}$$

Assuming the diode and the inductor to be ideal, there is no average power absorbed by either. The power supplied by the ac source is equal to the sum of the power absorbed by the resistor and the dc source

$$P_{\text{ac}} = I_{\text{rms}}^2 R + I_o V_{\text{dc}} \tag{3-27}$$

or it can be computed from

$$P_{\text{ac}} = \frac{1}{2\pi}\int_{0}^{2\pi} v(\omega t)\, i(\omega t)\, d(\omega t) = \frac{1}{2\pi}\int_{\alpha}^{\beta} (V_m \sin \omega t)\, i(\omega t)\, d(\omega t) \tag{3-28}$$

EXAMPLE 3-5

Half-Wave Rectifier with *RL*-Source Load

For the circuit of Fig. 3-5*a*, $R = 2\ \Omega$, $L = 20$ mH, and $V_{dc} = 100$ V. The ac source is 120 V rms at 60 Hz. Determine (*a*) an expression for the current in the circuit, (*b*) the power absorbed by the resistor, (*c*) the power absorbed by the dc source, and (*d*) the power supplied by the ac source and the power factor of the circuit.

■ Solution

From the parameters given,

$$V_m = 120\sqrt{2} = 169.7 \text{ V}$$
$$Z = [R^2 + (\omega L)^2]^{0.5} = 7.80\ \Omega$$
$$\theta = \tan^{-1}(\omega L/R) = 1.31 \text{ rad}$$
$$\alpha = \sin^{-1}(100/169.7) = 36.1^\circ = 0.630 \text{ rad}$$
$$\omega\tau = 377(0.02/2) = 3.77 \text{ rad}$$

(*a*) Using Eq. (3-22),

$$i(\omega t) = 21.8 \sin(\omega t - 1.31) - 50 + 75.3e^{-\omega t/3.77} \quad \text{A}$$

The extinction angle β is found from the solution of

$$i(\beta) = 21.8 \sin(\beta - 1.31) - 50 + 75.3e^{-\beta/3.77} = 0$$

which results in $\beta = 3.37$ rad (193°) using root-finding software.

(*b*) Using the preceding expression for $i(\omega t)$ in Eq. (3-24) and using a numerical integration program, the rms current is

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{0.63}^{3.37} i^2(\omega t)\,d(\omega t)} = 3.98 \text{ A}$$

resulting in

$$P_R = I_{\text{rms}}^2 R = 3.98^2(2) = 31.7 \text{ W}$$

(*c*) The power absorbed by the dc source is $I_o V_{dc}$. Using Eq. (3-26),

$$I_o = \frac{1}{2\pi}\int_{0.63}^{3.37} i(\omega t)\,d(\omega t) = 2.25 \text{ A}$$

yielding

$$P_{dc} = I_o V_{dc} = (2.25)(100) = 225 \text{ W}$$

(*d*) The power supplied by the ac source is the sum of the powers absorbed by the load.

$$P_s = P_R + P_{dc} = 31.2 + 225 = 256 \text{ W}$$

The power factor is

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{\text{rms}}} = \frac{256}{(120)(3.98)} = 0.54$$

■ PSpice Solution

The power quantities in this example can be determined from a PSpice simulation of this circuit. The circuit of Fig. 3-5*a* is created using VSIN, Dbreak, *R*, and *L*. In the simulation settings, choose Time Domain (transient) for the analysis type, and set the Run Time to 16.67 ms for one period of the source. Set the Maximum Step Size to 10 μs to get adequate sampling of the waveforms. A transient analysis with a run time of 16.67 ms (one period for 60 Hz) and a maximum step size of 10 μs is used for the simulation settings.

Average power absorbed by the 2-Ω resistor can be computed in Probe from the basic definition of the average of $p(t)$ by entering AVG(W(R1)), resulting in 29.7 W, or from $I_{\text{rms}}^2 R$ by entering RMS(I(R1))*RMS(I(R1))*2. The average power absorbed by the dc source is computed from the Probe expression AVG(W(Vdc)), yielding 217 W.

The PSpice values differ slightly from the values obtained analytically because of the diode model. However, the default diode is more realistic than the ideal diode in predicting actual circuit performance.